3.2 Induction on Numbers and on Formulas

Section 3.1 made formulas into objects: strings built from atoms and $\bot$ by five clauses, and by nothing else. There are infinitely many of them, and the theorems this chapter is heading for are about all of them. This section supplies the tool for proving anything about infinitely many objects at once. It is **induction**, first on the natural numbers, where it is familiar, then on formulas, where it takes the form this chapter uses again and again. With it we prove that every formula can be read in only one way, and that a definition given one clause per connective defines exactly one thing.

3.2.1 One argument for infinitely many cases

Here is a claim about every formula: *every formula has as many left parentheses as right ones.* It is believable. The clause (F3) of Definition 3.1.1 adds one of each, and the other clauses add none. But the claim is about infinitely many strings, and checking them one by one never ends.

The belief already contains the shape of a proof. Every formula is reached by finitely many building steps from atoms and $\bot$ (clause (F4)). The starting pieces have no parentheses, so they have equally many of each kind. Each building step turns formulas that balance into a formula that balances. So, step by step along any formation sequence, balance is never lost, and every formula balances.

Exactly the same shape appears with numbers. To show that some property holds for every natural number, show that it holds for 0, and that it passes from each number to the next. Every natural number is reached from 0 by finitely many steps of "add one", so the property is never lost. This is **induction**. We state it for numbers first, because it is simpler there, then prove the version for formulas from it.

Two words about language before we start. Statements such as "for every $n \in \mathbb{N}$, ..." and "there is a k with ..." are used in this section in their everyday sense, as Chapters 1 and 2 used them. Their rules of proof are the subject of Chapter 4, and nothing here needs more than the everyday reading. Likewise, a "collection" of formulas or numbers is meant in its everyday sense; *Sets, Relations, and Cardinality*, Chapter 1 treats sets properly. As in the rest of the book, $\mathbb{N} = \{0, 1, 2, \dots\}$ starts at 0, and the arithmetic and order of the natural numbers are used as school facts.

3.2.2 The principle of induction

A **property of natural numbers** is an open sentence $P(n)$ about a natural number n, such as "$n^2 \geq n$" or "$2^n > n$": for each particular n it becomes a statement, true or false (compare the open sentence $x > 2$ of Example (Statements and nonstatements), which became a statement once x was fixed).

If a property holds for 0*, and passes from each number to the next, then it holds for every number.*

Axiom 3.2.1. Principle of Induction

Let $P(n)$ be a property of natural numbers. Suppose that

(i) $P(0)$ holds, and

(ii) **for every** $n \in \mathbb{N}$, if $P(n)$ holds, then $P(n+1)$ holds.

Then $P(n)$ holds **for every** $n \in \mathbb{N}$.

In symbols (the quantifier symbols are Chapter 4's; here they abbreviate the words):

$$\Big(P(0) \ \wedge \ \forall n \in \mathbb{N}\ \big(P(n) \to P(n+1)\big)\Big) \ \to \ \forall n \in \mathbb{N}\ P(n).$$

In words: hypothesis (i) is the **base case**. Hypothesis (ii) is the **inductive step**, and inside it, the assumption "$P(n)$ holds" is the **induction hypothesis**. The step is **one** implication, proved **once**, for a number n about which we assume nothing except that it is a natural number with $P(n)$. The conclusion is about every natural number at once.

Its status. We take Axiom 3.2.1 as a basic property of $\mathbb{N}$, given and not proved, in the same way that *Analysis on the Real Line*, Chapter 1 takes the least upper bound axiom as a basic property of $\mathbb{R}$. It is, in a precise sense, what the three dots in $\{0, 1, 2, \dots\}$ mean: the natural numbers are what we reach from 0 by adding one, again and again, and nothing else. *Sets, Relations, and Cardinality*, Chapter 2 proves that it is equivalent to the well-ordering principle (every nonempty collection of natural numbers has a least element), and *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Chapter 1 derives it from the axioms of set theory. Nothing in this book depends on those two routes; every later use cites this principle.

Dominoes. The usual picture is a row of dominoes, one for each natural number. The base case knocks over the first. The inductive step says that each domino, if it falls, knocks over the next. Then every domino falls. The picture carries the one point that matters: the step must work for **every** n, the first transition from 0 to 1 included. A single domino standing too far from the next stops the whole row, however well the others are placed.

Not an infinite derivation. From $P(0)$ and the step we could derive $P(1)$ by $\to$E, then $P(2)$, and so on; each single $P(n)$ has a finite derivation, n lines long or so. But no finite derivation contains all of them, and "and so on" is not a rule. The principle of induction is what licenses the passage, in one step, from the two hypotheses to "for every n". It is a rule about $\mathbb{N}$, not a rule of logic.

The step as a box. In the language of Chapter 1, the inductive step is proved by a $\to$I box. We fix a natural number n, knowing nothing about it except what the step allows, and open a box assuming $P(n)$; inside the box we reach $P(n+1)$; closing the box gives the implication.

i	$P(n)$	
	$\vdots$	
j	$P(n+1)$	
k	$P(n) \to P(n+1)$	$\to$I i–j

The passage from "this n was arbitrary" to "for every n" is a rule for "for every", and Chapter 4 states it. For now, "let $n \in \mathbb{N}$" means what Chapter 1 made it mean (Section 1.1): one fixed number, not further specified.

Starting from another number. Often a property is claimed only from some $b \in \mathbb{N}$ on, as in "for every $n \geq 1$". Then it suffices to prove $P(b)$, and, for every $n \in \mathbb{N}$ with $n \geq b$, that $P(n)$ implies $P(n+1)$. Indeed, apply Axiom 3.2.1 to the property $Q(m)$: "$P(b+m)$ holds". Its base case $Q(0)$ is $P(b)$, and its step from $Q(m)$ to $Q(m+1)$ is the step from $P(b+m)$ to $P(b+m+1)$, where $b+m \geq b$. So $Q(m)$ holds for every $m \in \mathbb{N}$, and every natural number $n \geq b$ is $b+m$ for the natural number $m = n-b$. We call this induction **started at** b; it is Axiom 3.2.1 applied to Q.

Here is the first proof by induction, with every part labeled.

Example. A sum of products of neighbors

Prove that for every natural number $n \geq 1$,

$$1 \cdot 2 + 2 \cdot 3 + \cdots + n(n+1) = \frac{n(n+1)(n+2)}{3}.$$

Solution. Let $P(n)$ be the displayed equation. Its left side is the sum of the n terms $k(k+1)$ for $k = 1, \ldots, n$, so the left side for $n+1$ is the left side for n plus the one new term $(n+1)(n+2)$. Before proving anything, test small cases. For $n = 1, 2, 3$ the left side is 2, $2+6 = 8$ and $8+12 = 20$, and the right side is $\frac{1 \cdot 2 \cdot 3}{3} = 2$, $\frac{2 \cdot 3 \cdot 4}{3} = 8$ and $\frac{3 \cdot 4 \cdot 5}{3} = 20$. The claim is plausible. It starts at $n = 1$, so we use induction started at 1 (see above).

Base case. For $n = 1$ the left side is the single term $1 \cdot 2 = 2$, and the right side is $\frac{1 \cdot 2 \cdot 3}{3} = 2$. So $P(1)$ holds.

Inductive step. Let $n \in \mathbb{N}$ with $n \geq 1$, and suppose $P(n)$ holds (the induction hypothesis). Then

$$\begin{aligned} 1 \cdot 2 + \cdots + n(n+1) + (n+1)(n+2) &= \frac{n(n+1)(n+2)}{3} + (n+1)(n+2) \\ &= \frac{(n+1)(n+2)(n+3)}{3}, \end{aligned}$$

where the first equality is the induction hypothesis and the second takes out the common factor $(n+1)(n+2)$, leaving $\frac{n}{3} + 1 = \frac{n+3}{3}$. The last expression is the right side of $P(n+1)$. So $P(n)$ implies $P(n+1)$.

Conclusion. By Axiom 3.2.1, started at 1, $P(n)$ holds for every natural number $n \geq 1$.

Notice where the induction hypothesis entered: exactly once, to replace the sum of the first n terms by a closed expression. A proof by induction that never uses its induction hypothesis is either not an induction or is hiding a gap.

The inductive step of an inequality usually has a second ingredient besides the hypothesis: a crude estimate that finishes the job.

Example. An inequality by induction

Prove that $3^n \geq 2n + 1$ for every $n \in \mathbb{N}$.

Solution. Let $P(n)$ be "$3^n \geq 2n + 1$".

Base case. $3^0 = 1 = 2 \cdot 0 + 1$, so $P(0)$ holds.

Inductive step. Let $n \in \mathbb{N}$ and suppose $3^n \geq 2n + 1$. Then

$$3^{n+1} = 3 \cdot 3^n \geq 3(2n+1) = 6n + 3 \geq 2n + 3 = 2(n+1) + 1,$$

where the first inequality multiplies the induction hypothesis by the positive number 3, and

the second uses $6n \geq 2n$, true because $n \geq 0$. So $P(n+1)$ holds.

Conclusion. By Axiom 3.2.1, $3^n \geq 2n+1$ for every $n \in \mathbb{N}$.

The estimate $6n+3 \geq 2n+3$ throws away a great deal. That is typical: the step needs only to reach $P(n+1)$, not to reach it tightly, and a generous estimate that is obviously true is better than a sharp one that needs its own proof.

Warning. The inductive step must hold for **every** n, including the first transition. Try to prove "$2^n \geq n^2$ for every $n \in \mathbb{N}$": the base case $1 \geq 0$ holds, and the step $2^{n+1} = 2 \cdot 2^n \geq 2n^2 \geq (n+1)^2$ looks fine, but its last inequality holds only for $n \geq 3$, so the steps from 0 to 1, from 1 to 2 and from 2 to 3 are not covered. The claim survives the first two gaps (at $n = 1$, $2 \geq 1$, and at $n = 2$ both sides are 4), but it is false at $n = 3$, where $2^3 = 8 < 9 = 3^2$; it is true again from $n = 4$ on. *Sets, Relations, and Cardinality*, Chapter 2 shows a famous "proof" that fails in the same way at its very first step, from 1 to 2.

3.2.3 Strong induction

Sometimes the step from n to $n+1$ is the wrong shape. The immediate parts of a formula $(\varphi \circ \psi)$ are shorter than it, but not by exactly one symbol; a natural number may be best understood through a smaller one that is not its predecessor. In such cases we want to assume the property not just for n, but for **every** smaller number.

If a property holds for n whenever it holds for all smaller numbers, then it holds for every number.

Theorem 3.2.2. Strong Induction

Let $P(n)$ be a property of natural numbers. Suppose that **for every** $n \in \mathbb{N}$: if $P(k)$ holds for **every** $k \in \mathbb{N}$ with $k < n$, then $P(n)$ holds. Then $P(n)$ holds for every $n \in \mathbb{N}$.

This form is also called **course-of-values induction**: the hypothesis may use the whole course of values $P(0), \ldots, P(n-1)$, not just the last one. Its induction hypothesis is stronger (we assume more), so its step is easier to prove; the theorem says we lose nothing by it.

There seems to be no base case. There is one, hidden. For $n = 0$, the hypothesis says: if $P(k)$ holds for every $k \in \mathbb{N}$ with $k < 0$, then $P(0)$ holds. No natural number is less than 0, so "$P(k)$ holds for every $k \in \mathbb{N}$ with $k < 0$" is **vacuously** true: for each $k \in \mathbb{N}$, the implication "if $k < 0$, then $P(k)$" has a false hypothesis, and an implication with a false hypothesis holds (Proposition 2.1.2, $\neg P \vdash P \to Q$). So at $n = 0$ the hypothesis of the theorem hands us $P(0)$ outright. In practice, the proof of the step usually treats small n separately, and that is where the base cases live.

Idea. Strong induction is ordinary induction applied to a stronger property. Let $Q(n)$ say "$P(k)$ holds for **every** $k \leq n$". Then $Q(n)$ is exactly the hypothesis the theorem needs at $n+1$, since the numbers below $n+1$ are those up to n; so the theorem's hypothesis moves us from $Q(n)$ to $Q(n+1)$. Ordinary induction on Q does the rest, and $Q(n)$ contains $P(n)$.

Proof. Suppose the hypothesis of the theorem holds. For $n \in \mathbb{N}$, let $Q(n)$ be the property "$P(k)$ holds for every $k \in \mathbb{N}$ with $k \leq n$". We prove $Q(n)$ for every n by Axiom 3.2.1.

Base case. No natural number is less than 0, so $P(k)$ holds for every $k \in \mathbb{N}$ with $k < 0$, vacuously. By the hypothesis of the theorem at $n = 0$, $P(0)$ holds. The only $k \in \mathbb{N}$ with $k \leq 0$ is 0, so $Q(0)$ holds.

Inductive step. Let $n \in \mathbb{N}$ and suppose $Q(n)$. The natural numbers $k < n+1$ are exactly those with $k \leq n$, since no natural number lies strictly between n and $n+1$. So $Q(n)$ says that $P(k)$ holds for every $k \in \mathbb{N}$ with $k < n+1$, and the hypothesis of the theorem at $n+1$ gives $P(n+1)$. A natural number $k \leq n+1$ satisfies $k \leq n$ or $k = n+1$, and $P(k)$ holds in both cases. Hence $Q(n+1)$.

By Axiom 3.2.1, $Q(n)$ holds for every $n \in \mathbb{N}$. Given $n \in \mathbb{N}$, $Q(n)$ applied to $k = n$ gives $P(n)$. This proves the theorem.

■

Quick check. In the proof of Theorem 3.2.2, to which property is Axiom 3.2.1 applied, and why is it not applied to $P(n)$ itself?

Solution. To $Q(n)$: "$P(k)$ holds for every $k \in \mathbb{N}$ with $k \leq n$". Applied to $P(n)$ itself, ordinary induction would give us only $P(n)$ as induction hypothesis, while the hypothesis of the theorem at $n+1$ asks for $P(k)$ at **every** $k \leq n$. The property Q carries exactly that information from one step to the next.

The first use shows the typical shape: the step splits into a few small cases, handled directly, and a general case that reaches back further than one.

Example. Threes and fives

Prove that every natural number $n \geq 8$ is a sum of threes and fives: $n = 3a + 5b$ for some integers $a, b \geq 0$.

Solution. Ordinary induction is awkward here. From $n = 3a + 5b$ we cannot easily reach $n+1$: we would have to trade some threes for fives or the other way round, and there may be none of the kind we want to trade. Going back **three** steps is easy instead: if $n - 3$ is a sum of threes and fives, add one more three. So we use strong induction.

For $n \in \mathbb{N}$, let $P(n)$ be "if $n \geq 8$, then $n = 3a + 5b$ for some integers $a, b \geq 0$". Let $n \in \mathbb{N}$, and suppose $P(k)$ holds for every $k \in \mathbb{N}$ with $k < n$. We show $P(n)$.

Case $n < 8$. Then the hypothesis "$n \geq 8$" of $P(n)$ is false, and $P(n)$ holds vacuously.

Cases $n = 8, 9, 10$. Directly: $8 = 3 + 5$, $9 = 3 \cdot 3$, $10 = 5 \cdot 2$.

Case $n \geq 11$. Let $k = n - 3$. Then $k \geq 8$, so $k \in \mathbb{N}$, and $k < n$. By the induction hypothesis $P(k)$, there are integers $a, b \geq 0$ with $n - 3 = 3a + 5b$. Then $n = 3(a+1) + 5b$, with $a + 1 \geq 0$

and $b \geq 0$.

In every case $P(n)$ holds. By Theorem 3.2.2, $P(n)$ holds for every $n \in \mathbb{N}$: every $n \geq 8$ is a sum of threes and fives.

The three direct cases are the base cases. They are needed because the general case reaches back three steps: for $n = 8, 9, 10$, the number $n - 3$ is $5, 6, 7$, which is below 8, and there the induction hypothesis $P(n-3)$ is vacuous and gives us nothing. The bound 8 cannot be lowered: 7 is not a sum of threes and fives, since neither 7 nor $7 - 5 = 2$ is a multiple of 3.

3.2.4 Structural induction

We return to formulas. The formation sequence of a formula plays the part that "$0, 1, 2, \ldots$" plays for numbers: every formula is reached from atoms and $\bot$ by finitely many uses of the clauses. So a property of formulas can be proved in the same two parts.

A **property of formulas** is a statement $P(\varphi)$ about a formula φ, true or false for each particular formula: "φ has as many left as right parentheses", "φ contains the atom p".

If every atom and $\bot$ has a property, and each way of building a formula passes the property from the parts to the whole, then every formula has it.

Theorem 3.2.3. Structural Induction

Let $P(\varphi)$ be a property of formulas. Suppose that

(i) $P(p)$ holds for every atom p, and $P(\bot)$ holds;

(ii) **for every** formula φ, if $P(\varphi)$ holds, then $P(\neg\varphi)$ holds;

(iii) **for all** formulas φ, ψ and **each** binary connective $\circ$, if $P(\varphi)$ and $P(\psi)$ hold, then $P((\varphi \circ \psi))$ holds.

Then $P(\varphi)$ holds for every formula φ.

In words: (i) is the base case, one for each kind of formula of length one. (ii) and (iii) are the inductive steps, one for each building clause of Definition 3.1.1, and inside them "$P(\varphi)$ and $P(\psi)$" is the induction hypothesis, assumed for the **immediate parts** of the formula being built. There are six cases in all when the four binary connectives are counted separately; often the four are handled at once, with $\circ$ left unspecified.

Idea. Take any formula and a formation sequence ending in it. Walk along the sequence: each entry is an atom or $\bot$, which has the property by (i), or is built from **earlier** entries, which have it already, by (ii) or (iii). "Earlier entries, any of them" is exactly the shape of strong induction, applied to the position in the list.

Proof. Suppose (i)–(iii) hold, and let χ be a formula. By Definition 3.1.1 (F4) there is a formation sequence $\chi_1, \ldots, \chi_m$ with $\chi_m = \chi$. For $i \in \mathbb{N}$, let $Q(i)$ be the property "if $1 \leq i \leq m$, then $P(\chi_i)$ holds". We prove $Q(i)$ for every $i \in \mathbb{N}$ by Theorem 3.2.2.

Let $i \in \mathbb{N}$, and suppose $Q(k)$ holds for every $k \in \mathbb{N}$ with $k < i$. If $i = 0$ or $i > m$, then $Q(i)$ holds vacuously. So let $1 \leq i \leq m$. By the definition of a formation sequence, one of three

things is true.

- χ_i is an atom or $\bot$. Then $P(\chi_i)$ holds by (i).
- $\chi_i = \neg\chi_j$ for some $j < i$. Then $1 \leq j \leq m$, and $Q(j)$ gives $P(\chi_j)$. The entry χ_j is a formula (every entry of a formation sequence is one, Section 3.1), so (ii) applies and gives $P(\chi_i)$.
- $\chi_i = (\chi_j \circ \chi_l)$ for some $j, l < i$ and a binary connective $\circ$. Then $1 \leq j, l \leq m$, and $Q(j)$ and $Q(l)$ give $P(\chi_j)$ and $P(\chi_l)$, both entries are formulas, and (iii) gives $P(\chi_i)$.

So $Q(i)$ holds. By Theorem 3.2.2, $Q(i)$ holds for every $i \in \mathbb{N}$; in particular $Q(m)$ gives $P(\chi_m)$, that is, $P(\chi)$. Since χ was an arbitrary formula, this proves the theorem.

■

The proof uses **strong** induction, because an entry of a formation sequence may be built from any earlier entries, not only from the one just before it.

Remark. A tempting alternative is induction on the **height** of a formula: prove the property for all formulas of height 1, then for height $h+1$ assuming it for smaller heights. That route is circular here. The height was defined through "the" parse tree (Definition 3.1.4), and a formula has only one parse tree only because of the unique readability theorem below, whose proof uses structural induction. Inducting on the position in a formation sequence needs nothing but Definition 3.1.1.

Warning. A structural induction must cover **every** clause, the base case $\bot$ and the step for $\neg$ included. Try to prove "every formula contains an atom": it holds for atoms, and the steps for $\neg$ and for $(\varphi \circ \psi)$ keep it, since the atoms of the parts are atoms of the whole. But the case $\bot$ fails, and so does the claim: $\bot$, $\neg\bot$ and $(\bot \to \bot)$ contain no atom. A proof that "forgets" $\bot$ proves something false.

Quick check. Which clause of Definition 3.1.1 makes step (ii) of Theorem 3.2.3 necessary? Give a property of formulas that holds for every atom and for $\bot$, passes from φ and ψ to $(\varphi \circ \psi)$ for every binary connective, and yet fails for some formula.

Solution. Clause (F2), which builds $\neg\varphi$ from φ: step (ii) is the step for that clause. Without it, the formulas made by (F2) would be missed.

One such property: "φ is an atom, or is $\bot$, or begins with (". Atoms and $\bot$ have it, and every formula $(\varphi \circ \psi)$ begins with (, whatever its parts are. But $\neg p$ does not have it. So (i) and (iii) alone do not prove a property of every formula; the property fails exactly at a formula made by the forgotten clause.

3.2.5 Counting parentheses

Our first structural induction proves the claim this section began with, together with two refinements: one that the next theorem needs, and one that Exercise C1 uses. We count parentheses in strings, and for this some words about strings.

An **initial segment** of a string s is the string of the first k symbols of s, for some k with $0 \le k \le$ the length of s. For $k = 0$ it is the **empty** string, with no symbols; for k equal to the length it is s itself. An initial segment is **proper** if it is shorter than s. The **excess** of a string s is

$$\mathrm{ex}(s) := (\text{number of } (\text{ in } s) - (\text{number of }) \text{ in } s).$$

So a left parenthesis has excess 1, a right parenthesis -1, and any other symbol 0; and the excess of a string written as two strings one after the other is the sum of their excesses, since the counts add. For example, $\mathrm{ex}((p \wedge (q) = 2$: after reading $(p \wedge (q$ we are inside two pairs of parentheses that have not yet closed. That is the picture to keep: the excess of an initial segment is the **depth** we are at when we have read that far.

Lemma 3.2.4. Parenthesis Count

Let χ be a formula.

(a) $\mathrm{ex}(\chi) = 0$: χ has as many left as right parentheses.

(b) Every initial segment u of χ has $\mathrm{ex}(u) \ge 0$. If moreover χ has the form $(\varphi \circ \psi)$ and u is proper and nonempty, then $\mathrm{ex}(u) \ge 1$.

(c) For every occurrence of a binary connective in χ, the initial segment of χ that ends just before that occurrence has excess at least 1.

In words: (a) is balance. (b) says that reading a formula from the left, we never close more parentheses than we have opened, and that inside a formula $(\varphi \circ \psi)$ we stay at depth at least 1 until the very last symbol. (c) says that no binary connective lies outside all parentheses. Part (a) is the fact we set out to prove; (c), with (a), is what the unique readability theorem needs, and (b) shows that no proper initial segment of a formula is a formula (Exercise C1). The steps for (b) and (c) use (a) for the parts, so the three are proved together.

Idea. Induction on the shape of a formula, all three parts at once. For $\neg\varphi$, the extra $\neg$ changes no count. For $(\varphi \circ \psi)$, list its initial segments by where they stop: before anything, inside φ, at the connective, inside ψ, at the end. Each is "(" followed by pieces whose excess the induction hypothesis controls.

Proof. We prove, by Theorem 3.2.3, that every formula χ has the property $P(\chi)$: "(a), (b), and (c) hold for χ".

Atoms and $\bot$. The formula is one symbol, not a parenthesis. (a) Its excess is 0. (b) Its initial segments are the empty string and the formula itself, both of excess 0; and it does not have the form $(\varphi \circ \psi)$, which has length at least 5. (c) It contains no binary connective.

Negation. Let φ be a formula with $P(\varphi)$; we show $P(\neg\varphi)$. (a) $\mathrm{ex}(\neg\varphi) = 0 + \mathrm{ex}(\varphi) = 0$, by (a) for φ. (b) An initial segment of $\neg\varphi$ is either empty, of excess 0, or $\neg$ followed by an initial segment u' of φ, of excess $\mathrm{ex}(u') \ge 0$ by (b) for φ. The formula $\neg\varphi$ begins with $\neg$, so it does not have the form $(\varphi' \circ \psi')$, and the second sentence of (b) says nothing about it. (c)

A binary connective in $\neg\varphi$ lies in φ, and the initial segment of $\neg\varphi$ before it is $\neg$ followed by the initial segment of φ before it, whose excess is at least 1 by (c) for φ.

Binary connectives. Let φ, ψ be formulas with $P(\varphi)$ and $P(\psi)$, let $\circ$ be a binary connective, and let $\chi = (\varphi \circ \psi)$.

(a) $\mathrm{ex}(\chi) = 1 + \mathrm{ex}(\varphi) + 0 + \mathrm{ex}(\psi) - 1 = 0$, by (a) for φ and for ψ.

(b) Every initial segment of χ is of exactly one of the following five kinds, and its excess is as shown:

- the empty string: excess 0;
- (followed by an initial segment u' of φ: excess $1 + \mathrm{ex}(u') \geq 1$, by (b) for φ;
- $(\varphi \circ$: excess $1 + \mathrm{ex}(\varphi) + 0 = 1$, by (a) for φ;
- $(\varphi \circ$ followed by a nonempty initial segment v' of ψ: excess $1 + 0 + \mathrm{ex}(v') \geq 1$, by (a) for φ and (b) for ψ;
- χ itself: excess 0, by (a) for χ.

All five have excess at least 0. The proper nonempty ones are those of the middle three kinds, and they have excess at least 1.

(c) An occurrence of a binary connective in χ is the connective $\circ$ between φ and ψ, or lies in φ, or lies in ψ. Before $\circ$ stands $(\varphi$, of excess 1. Before an occurrence in φ stands (followed by the initial segment of φ before it, of excess at least $1 + 1 = 2$ by (c) for φ. Before an occurrence in ψ stands $(\varphi \circ$ followed by the initial segment of ψ before it, of excess at least $1 + 0 + 1 = 2$ by (a) for φ and (c) for ψ. In all three cases the excess is at least 1.

So $P(\chi)$ holds. By Theorem 3.2.3, every formula has the property P, which proves the lemma.

■

The proof of (c) gave more than (c) states, and the next proof uses the extra. In a formula $(\varphi \circ \psi)$, the initial segment before the connective $\circ$ has excess exactly 1, and the initial segment before **any other** binary connective has excess at least 2. In the language of depth: the main connective is the only binary connective at depth one. That is the recipe of Section 3.1 for finding the root of a parse tree.

The three parts were proved together for a reason. The second sentence of (b), alone, cannot be proved by structural induction. For $(\neg(p \wedge q) \wedge r)$, the initial segment $(\neg(p$ is (followed by the initial segment $\neg(p$ of the formula $\neg(p \wedge q)$, and its excess is $1 + \mathrm{ex}(\neg(p)$; to know that this is at least 1, we need $\mathrm{ex}(\neg(p) \geq 0$, a fact about an initial segment of a **negation**, which the second sentence of (b) does not cover. The first sentence of (b) does. A property that is too weak to pass through the induction must be strengthened until it does, and here the strengthening is (b)'s first sentence.

3.2.6 Unique readability

We can now prove what Section 3.1 promised: a formula can be built in only one way.

Theorem 3.2.5. Unique Readability

Let χ be a formula that is neither an atom nor $\bot$. Then **exactly one** of the following holds:

(i) $\chi = \neg\varphi$ for a formula φ;
(ii) $\chi = (\varphi \circ \psi)$ for formulas φ, ψ and a binary connective $\circ$.

In case (i), φ is **uniquely** determined by χ; in case (ii), $\circ$, φ and ψ are uniquely determined by χ.

In words: every compound formula has one main connective, and one way of being split into its immediate parts. This is the precise form of Chapter 1's claim that a statement built from letters by connectives can be read in only one way. Existence is the easy half; the content is **uniqueness**. The danger it excludes is a string such as $p \wedge q \vee r$, which, if it were a formula, could be split at $\wedge$ or at $\vee$.

Idea. Some form holds, by looking at the last step (Section 3.1); the first symbol tells the two forms apart. For case (ii), suppose $\chi = (\varphi \circ \psi) = (\varphi' \circ' \psi')$. By the remark after Lemma 3.2.4, each main connective is the only binary connective at depth one; so both sit at the same position of the string, and everything else is forced.

Proof. By Section 3.1 (looking at the last step), χ is an atom, $\bot$, $\neg\varphi$ for a formula φ, or $(\varphi \circ \psi)$ for formulas φ, ψ; by hypothesis it is one of the last two. It cannot be both, since a string of the form $\neg\varphi$ begins with $\neg$ and one of the form $(\varphi \circ \psi)$ begins with (.

Case (i). If $\chi = \neg\varphi = \neg\varphi'$, removing the first symbol gives $\varphi = \varphi'$.

Case (ii). Let $\chi = (\varphi \circ \psi)$. Before the connective $\circ$ stands the initial segment $(\varphi$, of excess $1 + \mathrm{ex}(\varphi) = 1$ by Lemma 3.2.4 (a). Before a binary connective inside φ stands (followed by an initial segment of φ of excess at least 1, by Lemma 3.2.4 (c); so the excess is at least 2. Before a binary connective inside ψ stands $(\varphi \circ$ followed by an initial segment of ψ of excess at least 1, by (c) for ψ and (a) for φ; so again at least 2. Hence the position of $\circ$ in χ is **the only** position of a binary connective in χ whose initial segment before it has excess 1, and this description mentions only the string χ.

Now suppose also $\chi = (\varphi' \circ' \psi')$ for formulas φ', ψ' and a binary connective $\circ'$. By the same argument, the position of $\circ'$ is the only position with that description. So $\circ$ and $\circ'$ stand at the same position of χ, and $\circ = \circ'$. That position is the one just after (and φ, and also the one just after (and φ', so φ and φ' have the same length; both are the string of symbols of χ between the first symbol and that position, so $\varphi = \varphi'$. Likewise ψ and ψ' are both the string of symbols between that position and the last symbol, so $\psi = \psi'$. This proves the theorem. ∎

The theorem settles the question Section 3.1 left open about parse trees.

Every formula has exactly one parse tree. We know it has one. That it has only one is proved by structural induction. An atom or $\bot$ has only the one-node tree, since a tree made by (T2) or (T3) of Definition 3.1.2 is a tree for a string beginning with $\neg$ or (. A parse tree for $\neg\varphi$ must have been made by (T2), from a parse tree for some φ' with $\neg\varphi' = \neg\varphi$; so $\varphi' = \varphi$, and by the induction hypothesis the tree below the root is the one parse tree of φ. A parse tree for $(\varphi \circ \psi)$ must have been made by (T3), from parse trees for some φ', ψ' and a connective $\circ'$ with $(\varphi' \circ' \psi') = (\varphi \circ \psi)$; by Theorem 3.2.5, $\circ' = \circ$, $\varphi' = \varphi$ and $\psi' = \psi$, and by the induction hypothesis the two subtrees are the parse trees of φ and ψ. From now on we speak of **the** parse

tree of a formula, and the height of Definition 3.1.4 is a well-defined number.

3.2.7 Definition by recursion

Height, the set of atoms, the number of connectives: each is naturally defined **one clause per way of building a formula**. The height of $\neg\varphi$ is one more than that of φ; the atoms of $(\varphi \circ \psi)$ are those of φ together with those of ψ. Such a definition is called **recursive**: it defines the value at a formula through the values at its immediate parts. The question is whether it really defines something. For a string that could be split in two ways, the clauses might assign two different values, and then they would define nothing. Unique readability says this never happens.

We say **function on formulas** for a rule that assigns to each formula exactly one value; the values may be numbers, truth values, formulas, or collections of atoms. (*Sets, Relations, and Cardinality*, Chapter 1 makes "function" precise; nothing here needs more than the everyday sense.)

Give a value for each atom and for $\bot$, and say how to get the value of a compound formula from its parts; then exactly one function on formulas obeys these clauses.

Theorem 3.2.6. Structural Recursion

Suppose we are given

- a value a_p for each atom p, and a value $a_\bot$;
- a rule $g_\neg$ that produces a value $g_\neg(\varphi, x)$ from any formula φ and any value x;
- for each binary connective $\circ$, a rule $g_\circ$ that produces a value $g_\circ(\varphi, \psi, x, y)$ from any formulas φ, ψ and any values x, y.

Then there is **exactly one** function f on formulas such that, for every atom p, all formulas φ, ψ and every binary connective $\circ$,

$$\begin{aligned} f(p) &= a_p, \qquad f(\bot) = a_\bot, \\ f(\neg\varphi) &= g_\neg(\varphi, f(\varphi)), \\ f((\varphi \circ \psi)) &= g_\circ(\varphi, \psi, f(\varphi), f(\psi)). \end{aligned}$$

In words: the three lines are the **clauses** of a recursive definition, one for each clause of Definition 3.1.1. The rules $g_\neg$ and $g_\circ$ may look at the immediate parts themselves as well as at their values; most definitions use only the values, and the list of subformulas below uses the parts. The theorem says the clauses define a function (**existence**) and only one (**uniqueness**).

Idea. Build the function on short formulas first. A formula with one symbol is an atom or $\bot$, and the first clauses fix its value. A longer formula has immediate parts that are shorter, whose values are already fixed, and unique readability says which clause applies and to which parts, with no choice left. So the value at each formula is forced, length by length. Formally: an ordinary induction on n, proving that there is exactly one function on the formulas of length at most n that obeys the clauses.

Proof. For $n \in \mathbb{N}$ with $n \geq 1$, call a function h, defined on the formulas of length at most n, **n-good** if it obeys the three clauses at every formula of length at most n. This makes sense: the immediate parts of a formula are shorter than it, so the clauses at a formula of length

at most n mention values of h only at formulas of length at most n. Let $S(n)$ be "there is exactly one n-good function"; we prove $S(n)$ for every $n \geq 1$ by Axiom 3.2.1, started at 1.

Base case. By the first-symbol test (Section 3.1), the formulas of length 1 are the atoms and $\bot$. A 1-good function must give each atom p the value a_p and $\bot$ the value $a_\bot$, and the function that does so is 1-good, since the other two clauses concern no formula of length 1. So $S(1)$ holds.

Inductive step. Let $n \in \mathbb{N}$ with $n \geq 1$, suppose $S(n)$, and let h_n be the n-good function. Define h on the formulas of length at most $n+1$ as follows. At a formula of length at most n, let h agree with h_n. Let χ have length $n+1 \geq 2$; it is neither an atom nor $\bot$, so by Theorem 3.2.5 it is **either** $\neg\varphi$ for exactly one formula φ, **or** $(\varphi \circ \psi)$ for exactly one $\circ$, φ, ψ; the parts have length at most n. Let $h(\chi) := g_\neg(\varphi, h_n(\varphi))$ in the first case and $h(\chi) := g_\circ(\varphi, \psi, h_n(\varphi), h_n(\psi))$ in the second. Unique readability leaves no choice of case or of parts, so this is one value. The function h is $(n+1)$-good: at formulas of length at most n it agrees with the n-good h_n, including at their parts, and at formulas of length $n+1$ it obeys the clauses by construction.

For uniqueness, let h' be any $(n+1)$-good function. Its restriction to the formulas of length at most n is n-good, so it is h_n, by $S(n)$. At a formula $\chi = \neg\varphi$ of length $n+1$, the clause for $\neg$ gives $h'(\chi) = g_\neg(\varphi, h'(\varphi)) = g_\neg(\varphi, h_n(\varphi)) = h(\chi)$; at $\chi = (\varphi \circ \psi)$ the same argument with $g_\circ$ gives $h'(\chi) = h(\chi)$. So $h' = h$, and $S(n+1)$ holds. By Axiom 3.2.1, started at 1, $S(n)$ holds for every $n \geq 1$; let h_n denote the n-good function.

The function f. If $m \geq n$, the restriction of h_m to the formulas of length at most n is n-good, so it is h_n. For a formula χ of length n, let $f(\chi) := h_n(\chi)$; then $f(\chi) = h_m(\chi)$ for every $m \geq n$. The function f obeys the clauses: for $\chi = \neg\varphi$ of length m, $f(\chi) = h_m(\chi) = g_\neg(\varphi, h_m(\varphi)) = g_\neg(\varphi, f(\varphi))$, and likewise for the other two clauses. And it is the only one: a function f' on formulas obeying the clauses restricts, for each n, to an n-good function, which is h_n; so $f'(\chi) = h_n(\chi) = f(\chi)$ for a formula χ of length n. This proves the theorem.

■

Unique readability was used at one place: to define $h(\chi)$ by exactly one clause, applied to exactly one choice of parts. Without it, the clauses could demand two different values at one string, and then no function would obey them.

Example. Four recursive definitions

Each of the following is a function on formulas by Theorem 3.2.6.

- **Height:** $\mathrm{ht}(p) = \mathrm{ht}(\bot) = 1$, $\mathrm{ht}(\neg\varphi) = \mathrm{ht}(\varphi) + 1$, and
$$\mathrm{ht}((\varphi \circ \psi)) = 1 + \max\{\mathrm{ht}(\varphi), \mathrm{ht}(\psi)\}.$$
- **Atoms:** $\mathrm{atoms}(p) = \{p\}$; $\mathrm{atoms}(\bot)$ has no members; $\mathrm{atoms}(\neg\varphi) = \mathrm{atoms}(\varphi)$; and $\mathrm{atoms}((\varphi \circ \psi))$ consists of the members of $\mathrm{atoms}(\varphi)$ together with those of $\mathrm{atoms}(\psi)$.
- **Number of connectives:** 0 for atoms and $\bot$; one more than for φ at $\neg\varphi$; one more than the sum for φ and ψ at $(\varphi \circ \psi)$.
- **Subformulas:** the list of subformulas of p is p, and of $\bot$ is $\bot$; of $\neg\varphi$, the formula $\neg\varphi$ followed by the subformulas of φ; of $(\varphi \circ \psi)$, the formula $(\varphi \circ \psi)$ followed by the subformulas of φ and then those of ψ, each formula kept only at its first appearance.

Compute all four for $\chi = (\neg(p \wedge q) \to (q \vee \bot))$, and also its length.

Solution. Work up from the leaves. For the parts: $\mathrm{ht}(p \wedge q) = 1 + \max\{1,1\} = 2$, so $\mathrm{ht}(\neg(p \wedge q)) = 3$; $\mathrm{ht}(q \vee \bot) = 2$. Hence

$$\mathrm{ht}(\chi) = 1 + \max\{3,2\} = 4.$$

The atoms of $\neg(p \wedge q)$ are p and q, those of $q \vee \bot$ are just q ($\bot$ contributes none), so $\mathrm{atoms}(\chi) = \{p, q\}$. The connectives: $\wedge$ and $\neg$ on the left, $\vee$ on the right, $\to$ at the root, four in all. The subformulas, in the order the clause lists them: χ, $\neg(p \wedge q)$, $(p \wedge q)$, p, q, $(q \vee \bot)$, $\bot$; the second q is dropped, being already listed. The length is 14: the official formula consists of the fourteen symbols

$$(\ \neg\ (\ p\ \wedge\ q\ )\ \to\ (\ q\ \vee\ \bot\ )\ ).$$

Two checks connect these definitions with Section 3.1. The **recursive height agrees with the tree height**: the parse tree of $\neg\varphi$ is that of φ with one node on top, so each branch gains one node; the tree of $(\varphi \circ \psi)$ is a new root over the trees of φ and ψ, so a longest branch is the root followed by a longest branch of the taller of the two. A structural induction whose cases are these two sentences (and the one-node tree for atoms and $\bot$) shows the two heights are equal. In the same way, the recursive list of subformulas is the list of formulas at the nodes of the parse tree (Definition 3.1.3): the nodes of the tree of $\neg\varphi$ are its root, carrying $\neg\varphi$, and the nodes of the tree of φ.

The length of a formula also satisfies clauses of this shape, by counting symbols: a formula of length one for atoms and $\bot$, one more at $\neg\varphi$, and at $(\varphi \circ \psi)$ the lengths of φ and ψ plus 3 (two parentheses and the connective). For χ above: $(p \wedge q)$ has $1 + 1 + 3 = 5$ symbols, so $\neg(p \wedge q)$ has 6; $(q \vee \bot)$ has 5; so χ has $6 + 5 + 3 = 14$, as counted.

Warning. A recursive definition must have a clause for **every** way of building a formula, and each clause may use only the **immediate parts**. "Let $f(\varphi)$ be f of the longest subformula of φ plus one" is not a recursive definition: at φ itself the longest subformula is φ, and the "definition" refers to the value it is trying to define. The theorem applies only to clauses of the form it states.

3.2.8 Exercises

3.2.8.1 A. Check your understanding

Exercise. A1

(a) For each statement, say which kind of induction fits best (ordinary, strong, structural), and write down its base case(s) and the statement of its inductive step, without proving them: (i) $4^n - 1$ is divisible by 3 for every $n \in \mathbb{N}$; (ii) every natural number $n \geq 12$ is $4a + 5b$ for some integers $a, b \geq 0$; (iii) every formula contains an atom or $\bot$.

(b) True or false, with a reason: to apply Theorem 3.2.2, one must first check $P(0)$ separately.

(c) In the proof of Theorem 3.2.3, on what is the strong induction performed?

(d) Which parts of Lemma 3.2.4 does the proof of Theorem 3.2.5 use, and for which formulas?

(e) Where in the proof of Theorem 3.2.6 is Theorem 3.2.5 used, and why is it not needed for uniqueness?

Solution. (a) (i) Ordinary induction. Base case: $4^0 - 1 = 0$ is divisible by 3. Step: for every $n \in \mathbb{N}$, if $3 \mid 4^n - 1$, then $3 \mid 4^{n+1} - 1$. (It holds because $4^{n+1} - 1 = 4(4^n - 1) + 3$.)

(ii) Strong induction, reaching back four: from $n - 4 = 4a + 5b$ we get $n = 4(a + 1) + 5b$. With $P(n)$ "if $n \geq 12$, then $n = 4a + 5b$ for some integers $a, b \geq 0$", the step is: for every $n \in \mathbb{N}$, if $P(k)$ holds for every $k < n$, then $P(n)$. Inside it, $n < 12$ is vacuous, and the base cases are $n = 12, 13, 14, 15$: $12 = 4 \cdot 3$, $13 = 4 \cdot 2 + 5$, $14 = 4 + 5 \cdot 2$, $15 = 5 \cdot 3$. For $n \geq 16$, $k = n - 4 \geq 12$ and $k < n$, and the hypothesis $P(n - 4)$ is used.

(iii) Structural induction. Base cases: an atom contains itself, and $\bot$ contains $\bot$. Steps: if φ contains an atom or $\bot$, so does $\neg\varphi$; if φ and ψ do, so does $(\varphi \circ \psi)$, for each binary $\circ$. (Compare the warning in the section: with "an atom" alone, the case $\bot$ fails.)

(b) False. At $n = 0$ the hypothesis of Theorem 3.2.2 reads "if $P(k)$ holds for every $k \in \mathbb{N}$ with $k < 0$, then $P(0)$", and its "if" part holds vacuously, since no natural number is less than 0. So proving the hypothesis for every n includes proving $P(0)$; in practice small n are treated as separate cases inside the step, as in Example (Threes and fives).

(c) On the position i of an entry in a formation sequence $\chi_1, \ldots, \chi_m$ of the given formula, with the property "if $1 \leq i \leq m$, then χ_i has P". Not on the height, which at that point is not yet known to be well defined.

(d) Part (a) for φ, to get excess 1 before the main connective and to get excess $1+0$ before the part ψ; part (c) for φ and for ψ, to get excess at least 2 before every other binary connective. Here $\chi = (\varphi \circ \psi)$ is the formula being read. Part (b) is not used in that proof.

(e) In the inductive step, to define $h(\chi)$ for a formula χ of length $n + 1$: unique readability says which clause applies and to which parts, so the clauses give exactly one value. That two functions obeying the clauses agree needs no unique readability: by Theorem 3.2.3, if f and f' both obey the clauses, they agree at atoms and $\bot$; and if they agree at φ (and ψ), then at $\neg\varphi$ both equal $g_\neg(\varphi, f(\varphi))$, and at $(\varphi \circ \psi)$ both equal $g_\circ(\varphi, \psi, f(\varphi), f(\psi))$, whichever way the string may be split.

3.2.8.2 B. Practice

Exercise. B1

Prove that $2^n \geq n + 1$ for every $n \in \mathbb{N}$.

Solution. Let $P(n)$ be "$2^n \geq n + 1$".

Base case. $2^0 = 1 = 0 + 1$, so $P(0)$ holds.

Inductive step. Let $n \in \mathbb{N}$ and suppose $2^n \geq n + 1$. Then

$$2^{n+1} = 2 \cdot 2^n \geq 2(n + 1) = 2n + 2 \geq n + 2 = (n + 1) + 1,$$

where the first inequality multiplies the induction hypothesis by $2 > 0$, and the second uses $n \geq 0$. So $P(n+1)$ holds.

By Axiom 3.2.1, $2^n \geq n+1$ for every $n \in \mathbb{N}$.

Exercise. B2

Prove by structural induction that every formula has at most as many distinct subformulas as it has symbols. Use the recursive description of the subformulas in Example (Four recursive definitions) and the clauses for the length given after it.

Solution. For a formula φ, write $s(\varphi)$ for the number of distinct subformulas of φ and $\operatorname{len}(\varphi)$ for its length. We prove $s(\varphi) \leq \operatorname{len}(\varphi)$ by Theorem 3.2.3.

Atoms and $\bot$. The only subformula is the formula itself: $s = 1 = \operatorname{len}$.

Negation. Suppose $s(\varphi) \leq \operatorname{len}(\varphi)$. The subformulas of $\neg\varphi$ are $\neg\varphi$ and those of φ, so there are at most $1 + s(\varphi)$ of them (fewer if one repeats). Hence $s(\neg\varphi) \leq 1 + s(\varphi) \leq 1 + \operatorname{len}(\varphi) = \operatorname{len}(\neg\varphi)$.

Binary connectives. Suppose $s(\varphi) \leq \operatorname{len}(\varphi)$ and $s(\psi) \leq \operatorname{len}(\psi)$. The subformulas of $(\varphi \circ \psi)$ are the formula itself and those of φ and of ψ, so

$$s((\varphi \circ \psi)) \leq 1 + s(\varphi) + s(\psi) \leq 1 + \operatorname{len}(\varphi) + \operatorname{len}(\psi) < \operatorname{len}(\varphi) + \operatorname{len}(\psi) + 3,$$

and the right side is $\operatorname{len}((\varphi \circ \psi))$.

By Theorem 3.2.3, $s(\varphi) \leq \operatorname{len}(\varphi)$ for every formula φ. (The bound is far from tight: the parentheses never start a subformula of their own.)

Exercise. B3

The number of nodes of the parse tree of a formula is given by the recursion $\operatorname{nodes}(p) = \operatorname{nodes}(\bot) = 1$, $\operatorname{nodes}(\neg\varphi) = \operatorname{nodes}(\varphi) + 1$, $\operatorname{nodes}((\varphi \circ \psi)) = \operatorname{nodes}(\varphi) + \operatorname{nodes}(\psi) + 1$. Prove that every formula φ of height h satisfies $\operatorname{nodes}(\varphi) \leq 2^h - 1$, and give a formula of height 3 for which equality holds.

Solution. We prove $\operatorname{nodes}(\varphi) \leq 2^{\operatorname{ht}(\varphi)} - 1$ by Theorem 3.2.3, using the recursive height of Example (Four recursive definitions).

Atoms and $\bot$. $\operatorname{nodes} = 1 = 2^1 - 1$.

Negation. Let $h = \operatorname{ht}(\varphi)$ and suppose $\operatorname{nodes}(\varphi) \leq 2^h - 1$. Then $\operatorname{ht}(\neg\varphi) = h + 1$ and

$$\operatorname{nodes}(\neg\varphi) \leq 2^h - 1 + 1 = 2^h \leq 2^{h+1} - 1,$$

where the last step is $1 \leq 2^h$.

Binary connectives. Let $a = \operatorname{ht}(\varphi)$, $b = \operatorname{ht}(\psi)$, and suppose the bound for both. Then

$h := \mathrm{ht}((\varphi \circ \psi)) = 1 + \max\{a, b\}$, so $a \le h-1$ and $b \le h-1$, and $2^a, 2^b \le 2^{h-1}$. Hence

$$\mathrm{nodes}((\varphi \circ \psi)) \le (2^a - 1) + (2^b - 1) + 1 \le 2 \cdot 2^{h-1} - 1 = 2^h - 1.$$

By Theorem 3.2.3, the bound holds for every formula. Equality at height 3: $((p \wedge q) \wedge (r \wedge p))$ has $7 = 2^3 - 1$ nodes (a root, two $\wedge$ nodes, and four leaves). Equality needs every node above the bottom level to have two children.

3.2.8.3 C. Going deeper

Exercise. C1: Initial segments are not formulas

(a) Prove that no proper initial segment of a formula is a formula.

(b) Deduce, in a way different from the proof in the text, the uniqueness part of case (ii) of Theorem 3.2.5: if $(\varphi \circ \psi) = (\varphi' \circ' \psi')$ for formulas $\varphi, \psi, \varphi', \psi'$ and binary connectives $\circ, \circ'$, then $\varphi = \varphi'$, $\circ = \circ'$ and $\psi = \psi'$.

Solution (Hint 1). Structural induction on the formula whose initial segments you consider. Which part of Lemma 3.2.4 settles the case $(\varphi \circ \psi)$ at once?

Solution (Hint 2). For (b): remove the first symbol from both sides. Now both φ and φ' are initial segments of the same string.

Solution. (a) The empty string is not a formula, since every formula has at least one symbol. So it is enough to show that no proper **nonempty** initial segment of a formula χ is a formula. We prove this by Theorem 3.2.3.

Atoms and $\bot$. A formula of length 1 has no proper nonempty initial segment.

Negation. Suppose no proper nonempty initial segment of φ is a formula, and let u be a proper nonempty initial segment of $\neg\varphi$. Then u is $\neg$ followed by a proper initial segment u' of φ. Suppose, for a contradiction, that u is a formula. It begins with $\neg$, so it is not an atom or $\bot$ and not of the form $(\varphi'' \circ \psi'')$; by looking at the last step (Section 3.1), $u = \neg\varphi''$ for a formula φ'', and removing the first symbol gives $u' = \varphi''$. So u' is a formula, hence nonempty, and it is a proper nonempty initial segment of φ, contradicting the induction hypothesis.

Binary connectives. Let u be a proper nonempty initial segment of $(\varphi \circ \psi)$. By Lemma 3.2.4 (b), $\mathrm{ex}(u) \ge 1$. If u were a formula, Lemma 3.2.4 (a) would give $\mathrm{ex}(u) = 0$. So u is not a formula. (This case needs no induction hypothesis.)

By Theorem 3.2.3, no formula has a proper nonempty initial segment that is a formula, which proves (a).

(b) Suppose $(\varphi \circ \psi) = (\varphi' \circ' \psi')$. Removing the first symbol, the strings $\varphi \circ \psi)$ and $\varphi' \circ' \psi')$ are equal. Both φ and φ' are initial segments of this string, so the shorter of the two is an initial segment of the longer. If their lengths differed, the shorter would be a proper initial segment of the longer, and a formula, contradicting (a). So they have the same length, and being initial segments of the same string, $\varphi = \varphi'$. The next symbol is then $\circ$ on one side

and $\circ'$ on the other, so $\circ = \circ'$; what remains is $\psi) = \psi')$, and removing the last symbol gives $\psi = \psi'$.

Marking scheme. (10 points)

- 1: reduces (a) to nonempty segments (the empty string is not a formula).
- 2: sets up a structural induction with the property "no proper nonempty initial segment is a formula", with the atom and $\bot$ case.
- 3: negation case: reads the segment as $\neg$ followed by a proper initial segment of φ, and uses the last step of a formula beginning with $\neg$ to reach a contradiction with the hypothesis.
- 2: binary case: excess at least 1 by Lemma 3.2.4 (b) against excess 0 by (a).
- 2: (b): compares φ and φ' as initial segments of one string, concludes equal length by (a), then reads off $\circ$ and ψ.

Fatal errors (0 for the step): using Theorem 3.2.5 to prove (b); assuming the lengths of φ and φ' are equal.

Exercise. C2: Find the error

The following argument claims to prove that every formula contains an **even** number of parentheses.

(1) We prove, by strong induction on n, that every formula with n symbols has an even number of parentheses.

(2) Let $n \in \mathbb{N}$, let χ be a formula with n symbols, and suppose that every formula with fewer than n symbols has an even number of parentheses.

(3) If χ is an atom or $\bot$, it has no parentheses, and 0 is even.

(4) If $\chi = \neg\varphi$, then φ is a formula with $n-1$ symbols, so by hypothesis it has an even number of parentheses; and χ has the same parentheses as φ.

(5) If $\chi = (\varphi \circ \psi)$, remove the outer pair of parentheses. The string $\varphi \circ \psi$ has $n-2$ symbols, so by hypothesis it has an even number of parentheses.

(6) Putting the outer pair back adds 2, so χ has an even number of parentheses.

(7) By Theorem 3.2.2, every formula has an even number of parentheses.

(a) Is the conclusion true?

(b) Which line fails, and why?

(c) Show that the move made in that line is illegal by applying the same move to a different property that every formula has, and reaching a false statement.

(d) Repair the argument.

Solution. *(a)* Yes. By Lemma 3.2.4 (a), a formula has as many left as right parentheses, say k of each, so $2k$ in all.

(b) Line 5. The induction hypothesis of line 2 is about **formulas** with fewer than n symbols, and line 5 applies it to the string $\varphi \circ \psi$, which is not a formula: for $\chi = (p \wedge q)$, it is $p \wedge q$, which fails the first-symbol test of Section 3.1 (length 3, beginning with an atom). The hypothesis says nothing about such strings. Every other line is correct: line 4 applies the hypothesis to the formula φ, and lines 1, 2, 3, 6 and 7 are a correct frame for a strong induction on the

length.

(c) Take the property of Lemma 3.2.4 (c), that no binary connective lies outside all parentheses; every formula has it. Run the same argument for it: in the case $\chi = (\varphi \circ \psi)$, line 5's move applies the hypothesis to the string $\varphi \circ \psi$ and asserts that no binary connective of $\varphi \circ \psi$ lies outside all parentheses. For $\chi = (p \wedge q)$ this asserts it of $p \wedge q$, whose $\wedge$ lies outside all parentheses. So the move leads from true hypotheses to a false statement; it is not a valid step, even though in the original argument its conclusion happens to be true.

(d) In line 5, apply the hypothesis to the formulas φ and ψ, each with fewer than n symbols: they have $2a$ and $2b$ parentheses for some integers $a, b \geq 0$, and χ has $2a + 2b + 2$, which is even. (Alternatively, prove the claim by Theorem 3.2.3, with the same three cases.)

Exercise. C3: Exit check: structural induction

Say that a formula is a **$\neg\wedge$-formula** if its only connectives are $\neg$ and $\wedge$ and it does not contain $\bot$. Define, by Theorem 3.2.6, the number of nodes of the parse tree and the number of negations:

$$\begin{aligned} \mathrm{nodes}(p) &= 1, \\ \mathrm{nodes}(\neg\varphi) &= \mathrm{nodes}(\varphi) + 1, \\ \mathrm{nodes}((\varphi \wedge \psi)) &= \mathrm{nodes}(\varphi) + \mathrm{nodes}(\psi) + 1, \\ \mathrm{neg}(p) &= 0, \\ \mathrm{neg}(\neg\varphi) &= \mathrm{neg}(\varphi) + 1, \\ \mathrm{neg}((\varphi \wedge \psi)) &= \mathrm{neg}(\varphi) + \mathrm{neg}(\psi), \end{aligned}$$

with the clauses for $\bot$ and the other connectives chosen in any way (say, as for $\wedge$). Prove by structural induction that every $\neg\wedge$-formula φ satisfies $\mathrm{nodes}(\varphi) > \mathrm{neg}(\varphi)$.

Solution (Hint 1). Theorem 3.2.3 is about all formulas. Put the restriction to $\neg\wedge$-formulas inside the property.

Solution (Hint 2). Let $P(\varphi)$ be "if φ is a $\neg\wedge$-formula, then $\mathrm{nodes}(\varphi) > \mathrm{neg}(\varphi)$". The immediate parts of a $\neg\wedge$-formula are $\neg\wedge$-formulas.

Solution. For a formula φ, let $P(\varphi)$ be: "if φ is a $\neg\wedge$-formula, then $\mathrm{nodes}(\varphi) > \mathrm{neg}(\varphi)$". We prove $P(\varphi)$ for every formula φ by Theorem 3.2.3; for a $\neg\wedge$-formula it is the claim.

The immediate parts of a $\neg\wedge$-formula are $\neg\wedge$-formulas, since every symbol of a part is a symbol of the whole.

Atoms. $\mathrm{nodes}(p) = 1 > 0 = \mathrm{neg}(p)$.

$\bot$, and the connectives $\vee, \to, \leftrightarrow$. The formulas $\bot$ and $(\varphi \circ \psi)$ with $\circ$ one of $\vee, \to, \leftrightarrow$ are not $\neg\wedge$-formulas, so for them P holds vacuously, whatever the clauses chosen for them.

Negation. Suppose $P(\varphi)$, and let $\neg\varphi$ be a $\neg\wedge$-formula. Then φ is one, so $\mathrm{nodes}(\varphi) > \mathrm{neg}(\varphi)$ by $P(\varphi)$, and adding 1 to both sides gives $\mathrm{nodes}(\neg\varphi) > \mathrm{neg}(\neg\varphi)$.

Conjunction. Suppose $P(\varphi)$ and $P(\psi)$, and let $(\varphi \wedge \psi)$ be a $\neg\wedge$-formula. Then φ and ψ are, so $\text{nodes}(\varphi) > \text{neg}(\varphi)$ and $\text{nodes}(\psi) > \text{neg}(\psi)$. Adding the two inequalities and then 1,

$$\text{nodes}((\varphi \wedge \psi)) = \text{nodes}(\varphi) + \text{nodes}(\psi) + 1 > \text{neg}(\varphi) + \text{neg}(\psi) = \text{neg}((\varphi \wedge \psi)).$$

(Since the counts are integers, each hypothesis gives at least 1 to spare, so the left side exceeds the right by at least 3.)

By Theorem 3.2.3, $P(\varphi)$ holds for every formula, which proves the claim.

Marking scheme. (10 points)

- 2: states the property for every formula, in a form that covers exactly the $\neg\wedge$-formulas, with the recursive counts.
- 1: atom case.
- 2: negation case, from the hypothesis for φ, after checking that φ is a $\neg\wedge$-formula.
- 3: conjunction case, using the hypotheses for **both** parts.
- 2: names the induction principle and says why $\bot$ and the other connectives are harmless (vacuous).

Fatal errors (0 for the step): applying the hypothesis to the formula being proved; a conjunction case that uses one part only and says “similarly” for the other.